causing microbe. Pasteur found that an attenuated (weakened) form of a disease-causing germ, produced in a laboratory and injected into the host animal or human, was particularly effective at enabling the body's immune system to fight off the disease. At first, Pasteur faced strong opposition and alarm at the prospect, but he was able to develop vaccines for anthrax, fowl cholera, and rabies—the latter involving his first test on a human.

"Where observation is concerned, chance favors only the prepared mind."

Louis Pasteur

Annihilating germs

The focus later shifted to finding germ-killing agents, or antibiotics, such as penicillin—discovered by Alexander Fleming. A strategy of annihilating microbes has been followed ever since. Yet this "slash and burn" approach has drawbacks: it kills beneficial microbes as well as harmful ones, and also promotes resistance in bacteria that can then render antibiotics ineffective. Such antimicrobial resistance is a major global public health threat. Alongside humans, antimicrobials are used to prevent and treat infectious diseases in animals (including livestock) and plants.